Requirements Specification & Project Planning

Before starting any engineering or software development project, it is necessary to define the preliminary requirements that guide system implementation clearly. Well-defined requirements ensure that development activities remain aligned with project objectives while supporting structured planning and validation. Requirements are typically divided into functional and non-functional categories, collectively referred to as solution requirements [1].

This chapter outlines the operational expectations, system behaviours, and technical standards required for the successful implementation of the FURSAN Conjunction Risk Analysis and Optimisation Framework. It defines how the system processes satellite conjunction data, performs risk prediction, generates optimisation recommendations, and presents results through an integrated analytical dashboard. These requirements serve as the foundation for development and documentation throughout the project lifecycle.

Functional & Non-Functional Requirements

Defining functional and non-functional requirements establishes a clear understanding of system capabilities before design and development. These requirements ensure that the FURSAN framework achieves reliable collision risk analysis while maintaining performance, scalability, and usability.

Functional Requirements

Functional requirements describe the essential operations the system must perform and directly define the behaviour and capabilities of the framework [2]. They specify system inputs, processing mechanisms, and expected outputs [3].

The functional requirements for the FURSAN Conjunction Risk Analysis and Optimisation Framework are listed below.

- The system must ingest Conjunction Data Messages (CDMs) and parse orbital and covariance parameters.
- The framework must preprocess datasets through cleaning, normalisation, and feature engineering pipelines.
- The system must compute statistical features such as Mahalanobis distance and uncertainty metrics derived from covariance matrices.
- The machine learning module must predict collision risk levels using hybrid models combining Deep Neural Networks, LSTM architectures, and ensemble classifiers.
- The framework must classify conjunction events into predefined risk categories.
- The optimisation module must compute manoeuvre recommendations by solving a Quadratic Unconstrained Binary Optimisation (QUBO) formulation.
- The Simulated Annealing solver must generate delta-v manoeuvre solutions that minimise fuel cost while increasing miss distance.
- The system must validate manoeuvre trajectories using external astrodynamics simulation tools.
- The dashboard must visualise conjunction evolution, predicted risks, and optimisation outputs.
- Users must be able to upload datasets and obtain automated analytical reports.
- The system must support end-to-end processing from dataset upload to manoeuvre recommendation.

Functional Requirement	Significance
CDM data ingestion	Enables structured analysis of satellite conjunction events.
Feature engineering pipeline	Improves prediction accuracy and model learning capability.
Hybrid ML/NN prediction	Provides reliable collision risk estimation.
Risk classification	Supports prioritisation of high-risk events.
Optimisation solver	Produces actionable manoeuvre strategies.
Trajectory validation	Ensures operational feasibility of recommendations.
Dashboard visualization	Enhances the interpretability of analytical results.
Automated reporting	Supports decision-making and documentation.

Table 1. Significance of Functional Requirements

Non-functional Requirements

Non-functional requirements describe system qualities such as efficiency, usability, reliability, and scalability [1]. These attributes influence overall system performance and user experience [4].

- The system must process datasets efficiently with acceptable computation time.
- The framework must maintain consistent prediction performance across multiple experiments.
- The interface must be intuitive and accessible to both technical and non-technical users.
- The system must handle incomplete or noisy CDM datasets without failure.
- The architecture must be scalable to accommodate larger satellite catalogues and future real-time integration.
- The framework must ensure reproducibility of experiments through controlled pipelines.
- The dashboard must provide responsive visualisation updates.
- Error handling mechanisms must provide informative feedback to users.
- The system must maintain modularity to allow integration of future optimisation solvers or AI models

Non-Functional Requirement	Significance
Efficient processing	Enables practical analysis of large datasets.
Reliability	Ensures trustworthy predictions.
Usability	Improves accessibility and adoption.
Robust data handling	Prevents system failures from imperfect inputs.
Scalability	Supports long-term expansion.
Reproducibility	Ensures scientific validity.
Responsive visualization	Improves analytical interaction.
Modular design	Simplifies upgrades and maintenance.

Table 2. Significance of Non-Functional Requirements

Hardware and Software Requirements

Identifying required resources ensures that the FURSAN framework can be implemented effectively using suitable computational tools and environments.

Hardware Requirements

The hardware components required for the project focus on computational capability rather than embedded systems:

- Development workstation capable of machine learning model training and optimisation simulations.
- GPU acceleration for deep learning experiments.
- Secure storage for datasets, trained models, and experimental outputs.
- Internet connectivity for accessing datasets, repositories, and simulation tools.

Software Requirements

The following software tools support the development and validation of the FURSAN framework:

- Python 3.10 Environment
- TensorFlow/Keras
- Scikit-Learn / XGBoost
- Pandas & NumPy
- NASA General Mission Analysis Tool (GMAT)
- MATLAB (Academic License)
- Streamlit / Flask
- Plotly / Matplotlib
- Next.js
- TypeScript
- Node.js
- HTML / CSS / Tailwind CSS

User Interface Requirements

The User Interface (UI) defines how users interact with the FURSAN analytical system. The dashboard is designed to present complex orbital analytics through intuitive visualisations while maintaining accessibility.

- Simple navigation allowing users to upload datasets and execute analysis workflows.
- Dashboard panels displaying predicted risk levels and optimisation outputs.
- 3D orbital visualisations representing conjunction geometry.
- Event evolution tracking showing risk changes across multiple CDMs.
- Automated report generation for analysis summaries.
- Responsive layout compatible with different display resolutions.
- Consistent visual theme across all dashboard components.

Project Planning

Effective planning ensures systematic development and steady progress toward an end-to-end analytical framework. The FURSAN project follows an iterative sprint-based structure to incrementally build system components while validating performance at each stage.

Project Timeline

Effective project planning is required to ensure that all phases of development, from conceptualisation to final deployment, are completed within defined time and resource constraints. A detailed project plan was created for the FURSAN Conjunction Risk Analysis and Optimisation Framework to track development activities, identify dependencies between machine learning and optimisation components, allocate testing periods, and anticipate potential technical bottlenecks.

This planning approach organised the development lifecycle while maintaining continuous progress through iterative delivery aligned with Agile Scrum principles. Incremental milestones ensured that each subsystem, including data processing, artificial intelligence modelling, optimisation, and dashboard integration, was validated before progressing to subsequent stages.

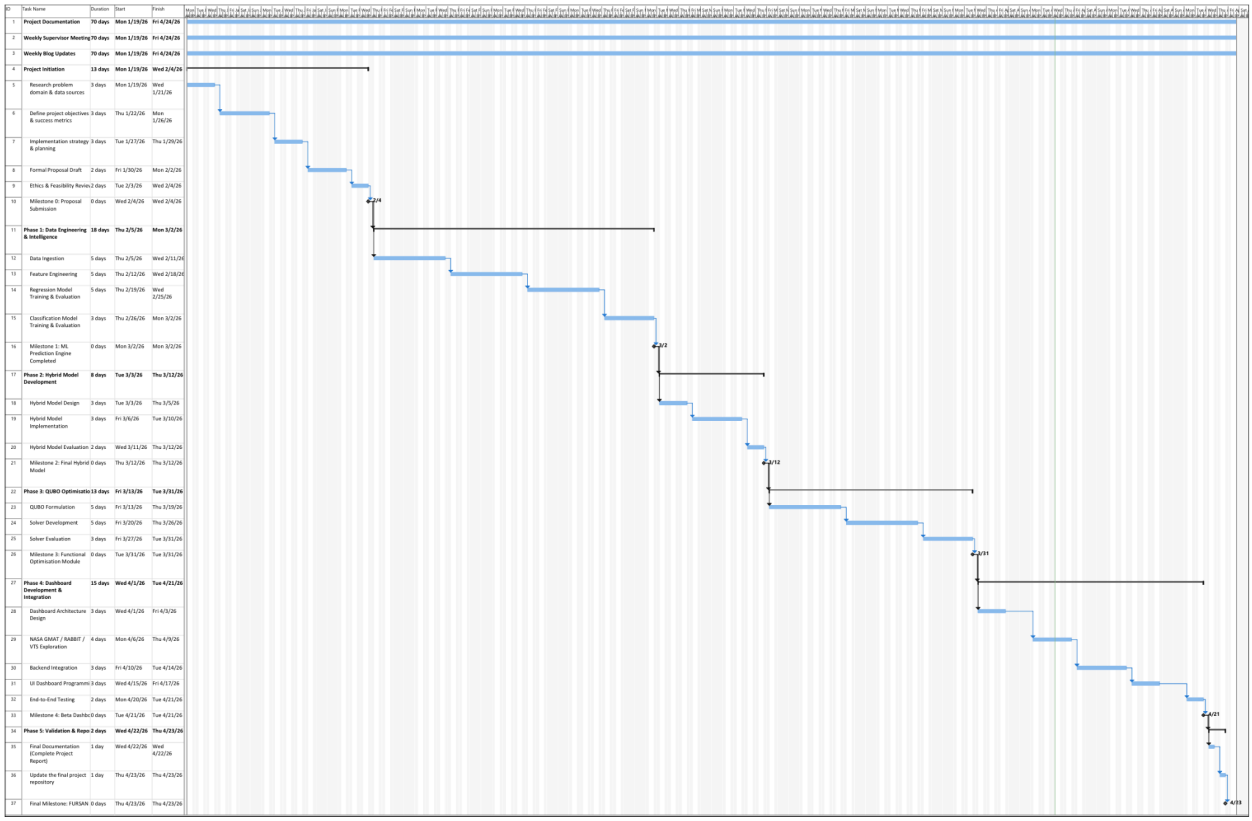
The project was broken down into five major phases:

1. Project Initiation
2. Design
3. Development
4. Testing
5. Deployment

Each phase includes defined objectives and deliverables contributing to the successful implementation of the FURSAN framework. Supporting activities such as documentation updates, supervisor meetings, experiment logging, and repository maintenance were conducted concurrently throughout the project to ensure transparency and consistent progress monitoring.

Gantt Chart

A Gantt Chart was designed to visualise the schedule and task overlaps for the FURSAN project. This project management tool maps each task with its start and end dates, durations in days, and dependencies. Each bar in the chart represents a specific task, while parallel bars indicate concurrent activities such as data processing, model training, optimisation development, dashboard integration, documentation, and supervisory meetings. The Gantt Chart allows easy tracking of task progress, understanding dependencies, and monitoring sequential and simultaneous activities.



ID	Task Name	Duration	Start	Finish
1	**Project Documentation**	**70 days**	**Mon 1/19/26**	**Fri 4/24/26**
2	**Weekly Supervisor Meeting**	**70 days**	**Mon 1/19/26**	**Fri 4/24/26**
3	**Weekly Blog Updates**	**70 days**	**Mon 1/19/26**	**Fri 4/24/26**
4	**Project Initiation**	**13 days**	**Mon 1/19/26**	**Wed 2/4/26**
5	Reasearch problem domain & data sources	3 days	Mon 1/19/26	Wed 1/21/26
6	Define project objectives & success metrics	3 days	Thu 1/22/26	Mon 1/26/26
7	Implementation strategy & planning	3 days	Tue 1/27/26	Thu 1/29/26
8	Formal Proposal Draft	2 days	Fri 1/30/26	Mon 2/2/26
9	Ethics & Feasibility Revie	2 days	Tue 2/3/26	Wed 2/4/26
10	Milestone 0: Proposal Submission	0 days	Wed 2/4/26	Wed 2/4/26
11	**Phase 1: Data Engineering & Intelligence**	**18 days**	**Thu 2/5/26**	**Mon 3/2/26**
12	Data Ingestion	5 days	Thu 2/5/26	Wed 2/11/26
13	Feature Engineering	5 days	Thu 2/12/26	Wed 2/18/26
14	Regression Model Training & Evaluation	5 days	Thu 2/19/26	Wed 2/25/26
15	Classification Model Training & Evaluation	3 days	Thu 2/26/26	Mon 3/2/26
16	Milestone 1: ML Prediction Engine Completed	0 days	Mon 3/2/26	Mon 3/2/26
17	**Phase 2: Hybrid Model Development**	**8 days**	**Tue 3/3/26**	**Thu 3/12/26**
18	Hybrid Model Design	3 days	Tue 3/3/26	Thu 3/5/26
19	Hybrid Model Implementation	3 days	Fri 3/6/26	Tue 3/10/26
20	Hybrid Model Evaluation	2 days	Wed 3/11/26	Thu 3/12/26
21	Milestone 2: Final Hybrid Model	0 days	Thu 3/12/26	Thu 3/12/26
22	**Phase 3: QUBO Optimisatio**	**13 days**	**Fri 3/13/26**	**Tue 3/31/26**
23	QUBO Formulation	5 days	Fri 3/13/26	Thu 3/19/26
24	Solver Development	5 days	Fri 3/20/26	Thu 3/26/26
25	Solver Evaluation	3 days	Fri 3/27/26	Tue 3/31/26
26	Milestone 3: Functional Optimisation Module	0 days	Tue 3/31/26	Tue 3/31/26
27	**Phase 4: Dashboard Development & Integration**	**15 days**	**Wed 4/1/26**	**Tue 4/21/26**
28	Dashboard Architecture Design	3 days	Wed 4/1/26	Fri 4/3/26
29	NASA GMAT / RABBIT / VTS Exploration	4 days	Mon 4/6/26	Thu 4/9/26
30	Backend Integration	3 days	Fri 4/10/26	Tue 4/14/26
31	UI Dashboard Programm	3 days	Wed 4/15/26	Fri 4/17/26
32	End-to-End Testing	2 days	Mon 4/20/26	Tue 4/21/26
33	Milestone 4: Beta Dashbo	0 days	Tue 4/21/26	Tue 4/21/26
34	**Phase 5: Validation & Repo**	**2 days**	**Wed 4/22/26**	**Thu 4/23/26**
35	Final Documentation (Complete Project Report)	1 day	Wed 4/22/26	Wed 4/22/26
36	Update the final project repository	1 day	Thu 4/23/26	Thu 4/23/26
37	Final Milestone: FURSAN	0 days	Thu 4/23/26	Thu 4/23/26

Figure 1: Gantt Chart for the FURSAN Project

The Gantt Chart provides a clear visual representation of the project phases, task durations, and overlapping activities. It serves as a useful reference to monitor critical milestones, track progress, and synchronise concurrent tasks such as weekly meetings, documentation, and experimentation logging.

Work Breakdown Structure

According to the Project Management Institute (PMI), a Work Breakdown Structure (WBS) is a hierarchical decomposition of the complete scope of work necessary to achieve the project's objectives and deliverables [6]. It is a core project management tool that allows organised planning, progress tracking, and resource allocation. A WBS visually organises tasks and subtasks, providing clarity on what needs to be done, who is responsible, and how individual activities contribute to the final system.

The WBS for the FURSAN project is organised into five main phases corresponding to the Gantt chart structure: Project Initiation, Data Engineering & Intelligence, Optimisation & Heuristics, Dashboard Development & Integration, and Validation & Reporting. Each phase is broken down into actionable tasks to allow effective monitoring, progress tracking, and milestone management.

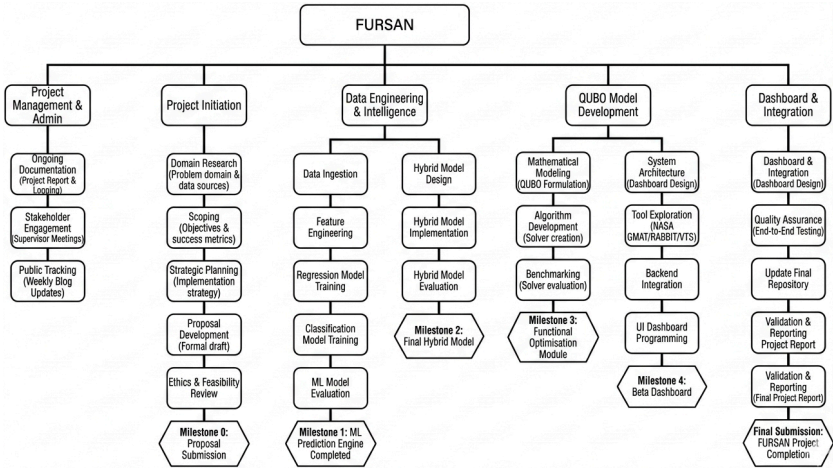


Figure 2:Work Breakdown Structure (WBS)

The WBS complements the Gantt chart by providing a clear breakdown of deliverables, ensuring each milestone is achievable and that no critical task is omitted. It facilitates delegation, risk identification, and milestone tracking, all essential for successfully managing a complex project like FURSAN.

References:

[1] A. Pandey, "Functional vs Non-Functional Requirements," GeeksforGeeks, 2023. [Online]. Available: https://www.geeksforgeeks.org/functional-vs-non functional-requirements/

[2] A. Ali, U. K. uz Zaman, and I. Ahmad, "Role of Non-functional Requirements in Projects' Success," in Proceedings of the International Conference on Software Engineering Research and Practices (SERP), 2022. [Online]. Available: https://www.researchgate.net/publication/361184121_Role_of_Non functional_Requirements_in_projects'_success

[3] N. Pitula, "Functional Requirements: The Basis of Successful Projects," Journal of Project Management and Innovation, vol. 10, no. 2, pp. 25–34, 2021.

[4] R. Singh and V. Singh, "Impact of Non-functional Requirements on Software Projects," International Journal of Software Engineering & Applications, vol. 13, no. 1, pp. 1–12, 2022.

[5] Project Management Institute, "What is Project Management?", 2023. [Online]. Available: https://www.pmi.org/about/learn-about-pmi/what-is-project management

[6] Project Management Institute, “Work Breakdown Structure (WBS): Basic Principles,” PMI, 2023. [Online]. Available: https://www.pmi.org/learning/library/work-breakdown-structure-basic principles-4883